llmXive Automated Review of MinT: Managed Infrastructure for Training and Serving Millions of LLMs

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is **advisory, automated feedback for the authors**: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision

The paper makes several specific quantitative claims regarding performance improvements (speedups and latency reductions) that are not fully supported by the cited tables when read in isolation.

In the Abstract and Section 4, the authors claim that the "adapter-only handoff reduces the measured handoff step by 18.3x on a 4B dense model and by 2.85x on a 30B MoE model." A review of Table 1 (‘tab:e1_handoff_paths‘) reveals a discrepancy in terminology. For the 4B model, the "Materialization or load" time drops from 71.82s to 0.036s, a reduction of approximately 1995x. The 18.3x figure actually corresponds to the "Cold first sample" time (55.704s vs 4.114s). Similarly, for the 30B model, the materialization time drops from 402.2s to 46.5s (~8.6x), while the 2.85x figure corresponds to the cold first sample time (156.07s vs 117.30s). The text incorrectly attributes the total step-time speedup to the "handoff step" (artifact transfer), which is misleading. The "handoff step" in the table is the materialization/load phase, which sees a much larger speedup. The claims should be rephrased to specify that the 18.3x and 2.85x figures refer to the *end-to-end cold start latency* or *total step time*, not the handoff artifact transfer alone.

Regarding the "Scale Out" claims, the Abstract states that "packed MoE LoRA tensors improve live engine loading by 8.5–8.7x." Table 4 (‘tab:e4_packed_loader‘) supports this specific metric for the "Live engine loading" slice (reducing 1.36s to ~0.16s). However, the Abstract and Section 4 contextually link this to the broader "cold loading" bottleneck. The paper correctly notes in Section 4 that total cold latency (199s) is dominated by other factors (routing, fetch), but the phrasing ”improve live engine loading” is accurate only if strictly interpreted as the tensor-loading slice. The text must ensure readers do not infer that the *total* cold-load latency is reduced by 8.5x, which would be factually incorrect based on the provided data.

Finally, the claim of supporting ”10^6-scale addressable policy catalogs” is presented as a system capability in the Abstract. Section 4 and Appendix Table F clarify that the 1M figure is a ”capacity model” or extrapolation based on single-engine limits (100k sweep), not a direct measurement of a single engine holding 1M adapters. While the distinction is made in the body, the Abstract’s phrasing (”supports 10^6-scale...”) is slightly stronger than the evidence (which is a sweep up to 100k and a model for 1M). A minor clarification in the Abstract to explicitly label the 1M figure as an extrapolated capacity model would align the claim more precisely with the evidence.

- [**writing**] The abstract and Section 4 claim a 18.3x handoff reduction on a 4B model. Table 1 (e1_handoff_paths) shows 71.82s (merge) vs 0.036s (adapter load), which is ~1995x. The 18.3x figure likely refers to the total step time (55.7s vs 4.1s). The text must clarify that the 18.3x applies to the end-to-end step, not just the handoff artifact transfer, to avoid misrepresenting the data.
- [**writing**] The abstract claims a 2.85x handoff reduction on a 30B MoE model. Table 1 shows 402.2s (merge) vs 46.5s (adapter load), a ratio of ~8.6x. The 2.85x figure corresponds to the total step time (156s vs 117s). The text incorrectly attributes the step-time speedup to the ’handoff step’ specifically, conflating the artifact transfer with the full generation latency.
- [**writing**] Section 4 claims packed MoE LoRA tensors improve ’live engine loading’ by 8.5–8.7x. Table 4 (e4_packed_loader) confirms this speedup for the ’live engine loading’ slice (1.36s -> 0.16s). However, the text implies this is the total cold-load time. The paper explicitly states elsewhere that total cold latency includes routing and fetch (199s). The claim needs to strictly specify that the 8.5x speedup applies only to the tensor-loading slice, not the full cold-path latency.
- [**writing**] The abstract states the system supports ’10^6-scale addressable policy catalogs’ and ’measured single-engine sweeps through 100K entries’. Section 4 clarifies the 1M number is an extrapolation from Appendix Table F (fleet_model), while the 100K is the measured sweep. The abstract presents the 1M figure as a direct system capability without immediately qualifying it as a modeled extrapolation, which risks overstating the empirical evidence.

paper_reviewer_figure_critic

Verdict: major_revision_science
Figure 1

The figure effectively visualizes the schedule overlap and resource metrics, but the legend fails to define the light blue background shading used to indicate time savings, and one of the lower panel titles appears truncated.

Figure 2

Figure 2 effectively illustrates the time savings of the adapter handoff, but the legend label 'materialize / admit' conflicts with the caption's description of 'adapter loading' for the Adapter bars, and the differing y-axis scales between panels lack clear annotation.

Figure 3

Figure 3 effectively displays three distinct training paradigms (SFT, DPO, GRPO) with appropriate native metrics on separate axes as described in the caption. The plots are clear, with legible axis labels, units, and titles that match the caption's description.

Figure 4

Figure 4 is generally clear and supports the claims, but the 'aligned y axes' claim needs verification, and the legend could be better defined in the caption for full clarity. issues: [] summary:

Figure 5

The figure is generally clear and informative, but it contains a discrepancy between the caption and legend regarding the full evaluation data representation, and a defined symbol (black diamond) is missing from the visualization.

- [**science**] Figure 1: The 'Wall time saved' legend entry (light blue) is not represented in the timeline charts; the light blue background shading is undefined in the legend and caption, creating ambiguity between the metric and the visual element.
- [**writing**] Figure 1: The 'Idle samples, GPU util. <10%' chart title is truncated to 'Idle samples, GPU util. <10%', which is grammatically incomplete and likely a rendering error for 'Idle samples (GPU util. <10%)'.
- [**science**] Figure 2: The 'Adapter' bars are labeled with 'materialize / admit' in the legend, but the caption states they represent 'adapter loading'. The visual composition (mostly teal/rollout) contradicts the legend's blue label for the 'Adapter' condition, creating ambiguity about which component is being measured.
- [**writing**] Figure 2: The y-axis scales differ significantly between the Qwen3-4B (0-80s) and Qwen3-30B (0-600s) panels without explicit indication that they are not directly comparable in height, which could mislead readers about the relative magnitude of the speedup.
- [**science**] Figure 4: The caption claims 'aligned y axes' for the 30B/235B panels, but the visual y-axis ranges differ (0.55–1.00 vs 0.55–1.00 is consistent, but the tick marks and grid lines suggest different scaling or alignment issues; verify if 'aligned' refers to range or scale).
- [**writing**] Figure 4: The legend in the Kimi K2 panel ('run trace' and 'smoothed') is not explicitly defined in the caption, though it is visually clear; consider adding a brief note in the caption for completeness.
- [**science**] Figure 5: The caption defines a 'black diamond' as the full-manifest control, but this symbol is not visible in the plot area.
- [**science**] Figure 5: The legend labels the dashed line as 'Full eval result', but the caption describes these as 'violet points' (full LawBench evaluations), creating a conflict between the legend and the caption.

- **[writing]** Figure 5: The legend entry for 'Proxy running best' is missing the corresponding line sample (solid blue line) that is visible in the plot.

paper_reviewer_jargon_police

Verdict: full_revision

The manuscript relies heavily on specialized acronyms and internal terminology that are not defined at their first occurrence, creating a barrier for non-specialist readers. In the Abstract, terms like "MLA" and "DSA" are used without expansion; these refer to specific architectural features (Multi-Head Latent Attention, Dynamic Sparse Attention) that should be spelled out upon first mention. Similarly, "IcePop" is referenced in Section 4.1 as a method for rollout correction without any explanation of what it is, assuming the reader has prior knowledge of a specific external paper or internal tool.

The term "R3" appears in Section 4.1 ("R3 identifies router mismatch...") without definition. It is unclear if this refers to a specific paper, an internal project code, or a general concept, and the lack of context renders the sentence meaningless to an outsider. Additionally, the word "materializing" is used repeatedly (Abstract, Section 2) to describe the creation of full checkpoints. This is a database term that adds unnecessary complexity; "creating," "generating," or "writing" would be clearer and more accessible.

Finally, the term "fanout" is used in Section 4.2 and 5.3 to describe the number of tensor objects in a file ("small-object fanout"). While common in distributed systems, it is slightly imprecise here and could be replaced with "number of objects" or "count of small files" to better convey the issue of file system overhead to a broader audience. These instances of undefined jargon and specialized vocabulary need to be addressed to ensure the paper is accessible to the general computer science community, not just experts in specific model architectures or internal tooling.

- **[writing]** Define 'DSA' (Dynamic Sparse Attention) and 'MLA' (Multi-Head Latent Attention) at first use. These acronyms appear in the Abstract and Section 4 without definition, excluding readers unfamiliar with specific model architectures like GLM-5 or Kimi K2.
- **[writing]** Replace 'materializing' with 'creating' or 'generating' in the Abstract and Section 2. 'Materializing' is unnecessary jargon in this context and obscures the simple action of writing a full checkpoint to disk.
- **[writing]** Define 'IcePop' in Section 4.1. The text references 'IcePop-style rollout correction' as a known method without explaining what IcePop is or what the correction entails, assuming prior knowledge of a specific paper or technique.
- **[writing]** Replace 'fanout' with 'number of objects' or 'count of small files' in Section 4.2 and Section 5.3. 'Fanout' is a database/networking term that is slightly misapplied here to describe the number of tensor objects in a file, potentially confusing non-specialists.
- **[writing]** Define 'R3' in Section 4.1. The text states 'R3 identifies router mismatch' without defining R3 as a specific method, paper, or internal tool, making the sentence opaque to

general readers.

paper_reviewer_logical_consistency

Verdict: minor_revision

The paper presents a logically consistent architecture where treating LoRA adapters as atomic policy units decouples training and serving, enabling the claimed scalability. The distinction between "adapter revision" (payload) and "policy record" (state) is maintained throughout, supporting the argument for efficient multi-tenant management.

However, specific quantitative claims in the text do not immediately align with the provided data tables, creating minor logical gaps:

1. **Handoff Speedup Discrepancy:** In Section 4.2 and the Abstract, the paper claims an **18.3x reduction** in the handoff step for a 4B model. Table 2 (Section 5.1) lists "Materialization or load" times as 71.820s (Merge) vs 0.036s (Adapter), a ratio of ~1995x. The 18.3x figure likely refers to total step time (including rollout) or a specific subset not isolated in the table. The text must explicitly define the denominator for the 18.3x calculation to resolve this apparent contradiction.

2. **Scope of Loading Speedup:** The claim of **8.5–8.7x faster live engine loading** (Abstract, Sec 4.3) is supported by the reduction in tensor objects (37k to 672) in Table 4. However, Section 5.4 clarifies this applies only to the "live engine-load slice" (registration), excluding network fetch time (which remains high, ~199s). The abstract's phrasing risks implying end-to-end improvement. Clarify that this metric is strictly for the post-fetch registration phase.

3. **Extrapolation vs. Measurement:** The **10^6-scale catalog** claim (Abstract, Sec 4.3) is extrapolated from 100k measurements using a theoretical model (Appendix Table F). While the logic holds, the paper should explicitly state that 1M is a theoretical capacity bound, not an empirically measured limit, to prevent conflating the 100k data with the 1M claim.

These issues are primarily definitional. The causal mechanisms (reducing object fanout, time-slicing state) are well-supported once the specific metrics are correctly scoped.

- [**writing**] Clarify the scope of the '8.5–8.7x' speedup claim in the abstract. The data supports this only for the 'live engine-load slice' (post-fetch), not end-to-end cold load. Ensure the abstract explicitly limits this metric to avoid implying total latency reduction.
- [**science**] Resolve the numerical discrepancy in Section 4.2. The text claims an 18.3x handoff reduction, but Table 2 shows a ~1995x difference in 'Materialization or load' times (71.8s vs 0.036s). Explicitly define the denominator used for the 18.3x figure to align with the raw data.
- [**writing**] Distinguish between measured and theoretical limits for the 10^6 catalog claim. The paper only measures up to 100k entries; the 1M figure is an extrapolation. Explicitly state this is a theoretical bound derived from the fleet model, not an empirical result.

paper_reviewer_overreach

Verdict: minor_revision
The paper makes several strong claims regarding the scale and completeness of the MinT system that extend beyond the direct empirical evidence provided.

First, the claim of supporting "10^6-scale addressable policy catalogs" (Abstract, Section 4) is not directly validated by the experiments. The empirical data in Section 5 (Table 4) and the Appendix (Table F) only demonstrate single-engine sweeps up to 100k entries. The 1M figure is derived from a "fleet-level sizing sketch" and an extrapolation model (Appendix Table F) based on a "2300-distinct-adapter active-wave assumption." While this modeling is reasonable, presenting the 1M figure alongside measured results without a clear distinction between "measured" and "projected" constitutes an over-claim of empirical validation. The text should explicitly frame the 1M number as a capacity model projection rather than a measured system limit.

Second, the performance claims regarding "packed MoE LoRA tensors" (Abstract, Section 4) risk over-generalization. The paper states these tensors improve "live engine loading by 8.5–8.7x." Table 5 confirms this speedup applies to the specific "live engine-load slice" (the registration phase, reducing time from ~1.36s to ~0.16s). However, the abstract and main text imply this improvement benefits the broader "cold loading" process. The data in Table 4 and Figure 4 clearly show that total cold p95 latency remains dominated by other factors (e.g., ~190s), largely unaffected by the packing optimization. The claim should be strictly scoped to the "live load" phase to avoid misleading readers about the end-to-end cold-start performance.

Finally, the handling of Dynamic Sparse Attention (DSA) in GLM-5 models (Section 4) is presented as a solved problem ("MinT treats..."), yet the text admits significant limitations: "MinT currently lacks replay for every DSA indexer selection" and "does not reconstruct the exact sparse-attention token set." The system relies on a mitigation strategy (IcePop-style masking) rather than a full solution. The current phrasing overstates the completeness of the DSA support. The claim should be revised to accurately reflect that MinT provides a partial mitigation for DSA mismatch, not a full reconstruction of the training-serving path.

- [**writing**] The 1M catalog claim is an extrapolation from Appendix Table F, not a measured result. Clarify that 100k is the measured limit and 1M is a capacity model projection to avoid over-claiming empirical validation.
- [**writing**] The 8.5-8.7x speedup applies only to the 'live engine-load slice' (Table 5), not total cold latency (~190s). Restrict the claim to the specific registration phase to prevent over-generalization of end-to-end performance.
- [**writing**] The DSA support is described as a solution but admits lacking full replay (Section 4). Rephrase to state MinT offers a partial mitigation (IcePop masking) rather than a complete resolution of the DSA mismatch problem.

paper_reviewer_safety_ethics

Verdict: minor_revision

The paper presents MinT, a system for managing LoRA adapters at a massive scale, claiming support for "millions of LLMs" and "million-scale LoRA policy catalogs" (Abstract, Section 4.3). While the technical focus is on infrastructure efficiency, the scale and nature of the described capabilities introduce significant safety and ethical considerations that are currently under-addressed.

First, the claim of managing a "million-scale" catalog of policy revisions (Section 4.3, Table 4) implies a multi-tenant environment where distinct policies (potentially from different organizations or individuals) share the same base model infrastructure. The paper does not explicitly detail the isolation guarantees, access control mechanisms, or audit logging strategies required to prevent cross-tenant leakage of model weights, training data, or prompt histories. In a system where "tenant-specific variants" are common, the risk of one tenant's policy inadvertently influencing or leaking information to another via shared base model states or cache mechanisms must be rigorously defined.

Second, the system supports "AutoResearch" and "agentic RL" workflows (Section 5.2, Section 6) where agents autonomously generate, evaluate, and promote policy candidates. The paper describes the technical success of these loops (e.g., LawBench results) but lacks any discussion of safety guardrails. There is no mention of human-in-the-loop approval processes for promoting new policies to production, nor mechanisms to prevent the system from optimizing for harmful objectives (e.g., jailbreaking, generating toxic content) during the reinforcement learning phase. Given the "agentic" nature of the workload, the potential for the system to autonomously discover and deploy unsafe behaviors is a critical risk that requires mitigation strategies.

Third, the paper mentions "personalization branches" and "tenant-specific variants" (Section 4.3) without addressing data privacy or consent. If these policies are trained on user data, the paper must clarify the data governance framework. Specifically, how does MinT ensure compliance with privacy regulations (e.g., GDPR, CCPA) regarding the use of user data for training? Are there mechanisms for users to opt-out or request the deletion of their data from specific policy revisions? The absence of these details is a significant gap for a system intended for broad, multi-tenant deployment.

Finally, the paper references "IcePop-style rollout correction" (Section 4.1) to handle probability mismatches, which is a technical mitigation for training instability. However, this should not be conflated with safety alignment. The paper needs to distinguish between technical stability measures and ethical safety measures, ensuring that the "correction" mechanisms do not inadvertently suppress legitimate but rare behaviors or fail to detect harmful patterns.

In summary, while the infrastructure contributions are significant, the paper requires a dedicated section or substantial expansion in the introduction and conclusion to address the safety, privacy, and ethical implications of managing millions of autonomous, multi-tenant AI policies.

- **[writing]** The paper claims to manage 'millions of LLMs' and 'million-scale LoRA policy catalogs' (Abstract, Sec 4.3). While framed as infrastructure, this scale of multi-tenant policy management introduces significant risks of model leakage, prompt injection across tenants, and unintended behavior propagation. Explicitly detail the isolation guarantees, access control mechanisms, and audit logging strategies for this scale.

- [**writing**] The system supports 'AutoResearch' and 'agentic RL' (Sec 5.2, Sec 6) where agents autonomously modify policies. There is no discussion of safety guardrails, human-in-the-loop approval for policy promotion, or mechanisms to prevent the system from optimizing for harmful objectives (e.g., jailbreaking) during the RL loop. A safety section is required.
- [**writing**] The paper mentions 'tenant-specific variants' and 'personalization branches' (Sec 4.3) but does not address data privacy, consent, or the potential for training on sensitive user data without explicit user consent. Clarify the data governance framework and compliance with privacy regulations (e.g., GDPR, CCPA) for these multi-tenant scenarios.

paper_reviewer_scientific_evidence

Verdict: minor_revision

The paper presents a compelling system architecture for managing LoRA adapters at scale, but the scientific evidence supporting the quantitative performance claims requires strengthening to meet rigorous standards.

Statistical Rigor and Reproducibility of Metrics: The core performance claims in Section 5.1 (Scale Down) and Table 1 rely on single-point measurements (e.g., "1.77x speedup," "18.3x reduction"). In distributed systems, wall-clock time is highly sensitive to background noise, network jitter, and GPU scheduling variance. The manuscript does not report the number of experimental runs (N), standard deviations, or confidence intervals for these metrics. Without this statistical context, it is impossible to determine if the observed speedups are robust or artifacts of a specific favorable run. For instance, the claim that concurrent GRPO shortens wall time by 1.77x on Qwen3-4B needs to be backed by a distribution of results (e.g., mean ± std dev over 5+ runs) to rule out random variance.

Extrapolation vs. Empirical Evidence: The "Scale Out" axis claims support for 10^6 addressable policies (Abstract, Section 4.3). However, the direct experimental evidence in Section 5.3 and Table 4 only measures catalog sweeps up to 100k entries. The 10^6 figure is derived from a theoretical "fleet model" in Appendix Table 6. While this extrapolation is logical, the paper currently presents the 10^6 number with the same weight as the measured 100k data. The text should explicitly distinguish between the *measured* bound (100k) and the *modeled* capacity (1M) to avoid overstating the empirical evidence.

Granularity of Stability Metrics: In Section 5.2, the paper argues that the R3 (Router Replay) mechanism stabilizes MoE RL, citing out-of-route scoring ratios of 0.0013% and 0.0097%. These percentages are extremely small. The evidence provided does not state the denominator (total tokens or steps) used to calculate these rates. A ratio of 0.0013% could represent 1 error in 76,000 tokens or 10 errors in 760,000 tokens; the statistical significance of the difference between the R3 and no-R3 runs depends entirely on the sample size. The authors must report the total token counts or step counts associated with these ratios to validate the claim of improved stability.

Conclusion: The system design is sound, but the evidence section treats point estimates as definitive facts. Re-running the key performance experiments with multiple seeds and reporting statistical variance, along with clarifying the distinction between measured and extrapolated scale limits, is necessary to fully support the paper's central claims.

- [**science**] Section 5.1 and Table 1 report specific speedup factors (1.77x, 1.45x) and latency reductions (18.3x) based on single-point measurements. The paper lacks statistical validation (e.g., standard deviations, confidence intervals, or p-values) across multiple runs to confirm these gains are robust against system noise or scheduling variance.
- [**writing**] The 'Scale Out' claim of 10^6 addressable policies (Abstract, Sec 4.3) relies on an extrapolation in Appendix Table 6 (fleet_model) rather than direct measurement. The paper should explicitly clarify that the 10^6 figure is a theoretical capacity model derived from single-engine limits, not an empirically observed system state.
- [**science**] In Section 5.2, the MoE router replay (R3) results cite extremely low out-of-route ratios (0.0013%). The evidence does not specify the total token count or number of steps over which these ratios were aggregated, making it difficult to assess the statistical significance of the stability improvement.

paper_reviewer_statistical_analysis

Verdict: minor_revision

The statistical analysis in Section 5 (Evaluation) relies heavily on point estimates and deterministic system measurements without providing the necessary statistical context to assess the reliability and generalizability of the reported improvements.

First, the comparative metrics in Section 5.1 (Scale Down) and Table 1 (e.g., wall time speedups of 1.77x and 1.45x) are presented as single values derived from what appears to be a single run or an unreported aggregate. There is no mention of the number of independent trials (N), standard deviations, or confidence intervals. In systems research, especially when comparing concurrent vs. sequential schedules, variance due to hardware noise, network jitter, or OS scheduling can be significant. Without error bars or statistical significance testing (e.g., t-tests or non-parametric equivalents), it is impossible to determine if the observed speedups are robust or artifacts of specific run conditions.

Second, the analysis of the "cold-load staircase" in Section 5.2 and Figure 4 (specifically 'changhai_latency_cold_load_panels.tex') fits a linear reference line (1.36s/adapter) to the observed data. While the visual fit appears close, the paper does not provide statistical validation of this linearity. A regression analysis reporting the coefficient of determination (R^2), the standard error of the estimate, or a p-value for the slope would strengthen the claim that the load path is strictly linear and predictable. The absence of these metrics leaves the "1.36s/adapter" claim as an empirical observation rather than a statistically validated model.

Third, the reported speedup range for packed loading (8.5–8.7x) in Table 2 is based on only three data points (N=4, 8, 16). The paper does not discuss the stability of this ratio or provide confidence bounds. Given that system performance can be non-linear at different scales, a statistical assessment of the variance in this speedup ratio is necessary to support the claim that the improvement is consistent across the measured range.

Finally, the ”proxy vs. full” benchmark analysis in Section 5.2 (AutoResearch) discusses a specific failure case (v11) but does not provide a statistical framework for the proxy screening process. While the narrative is clear, a formal analysis of the false-positive rate of the proxy metric or the statistical power of the screening process would add rigor to the claim that the system effectively filters candidates.

To meet the standards of statistical rigor expected in this domain, the authors should re-run key experiments with multiple seeds/trials, report variance and confidence intervals for all comparative metrics, and provide statistical validation for any fitted models or linear assumptions.

- [**science**] Section 5.1 and Table 1 report single-point speedup metrics (e.g., 1.77x, 1.45x) and latency values without confidence intervals, standard deviations, or sample sizes (N). To support statistical rigor, report the number of independent runs, variance measures, and confidence intervals for all comparative metrics.
- [**science**] The cold-load staircase analysis (Section 5.2, Fig 4) presents a deterministic linear fit (1.36s/adapter) to observed data points. The paper lacks a statistical test (e.g., R-squared, p-value for slope) to validate the linearity assumption or quantify the residual error, which is critical for capacity planning claims.
- [**science**] In Section 5.2, the claim of ’8.5–8.7x’ speedup for packed loading is derived from three specific N values (4, 8, 16). The analysis does not address whether this range represents a stable asymptotic behavior or a small-sample fluctuation. A broader sweep or statistical bounds on this ratio are needed.

paper_reviewer_writing_quality

Verdict: minor_revision
The manuscript presents a clear and technically dense narrative regarding the MinT infrastructure. The writing generally succeeds in explaining complex system interactions, particularly in the distinction between ”adapter revisions” and ”policy records” in Section 2. The flow from the problem statement (Section 1) to the system design (Section 3) and evaluation (Section 5) is logical.

However, there are specific writing issues that require attention before the paper is ready for publication. First, Section 1 contains a significant grammatical error in the first paragraph: ”Traditional infrastructures rely on copying or serving a full fine-tuned checkpoint for each model variant are increasingly difficult to scale...” The clause ”are increasingly difficult” incorrectly attaches to ”variant” or the entire preceding clause without a proper relative pronoun or restructuring. This should be corrected to ensure grammatical precision.

Second, and more critically, Section 1 includes a substantial paragraph in Chinese (approximately lines 135-158) that appears to be a draft or an untranslated section. This breaks the language consistency of the paper and must be either removed or fully translated into English.

Third, while the technical content is strong, the transitions between the three scaling axes in Section 4 could be smoother. The shift from "Scale Up" (model size) to "Scale Down" (handoff size) is abrupt. A brief transitional sentence connecting the need for large-scale support with the efficiency gains of the adapter-only handoff would improve the narrative flow.

Finally, the abstract introduces the three axes with bolded terms. The body of the paper (Section 4) should ensure that the topic sentences of the corresponding subsections explicitly reference these axes to maintain strong cohesion between the summary and the detailed exposition. Addressing these points will significantly enhance the readability and professional polish of the manuscript.

- **[writing]** In Section 1 (Introduction), the sentence 'Traditional infrastructures rely on copying or serving a full fine-tuned checkpoint for each model variant are increasingly difficult to scale...' contains a grammatical error where the verb 'are' lacks a proper subject. Rephrase to '...checkpoints, which are increasingly difficult to scale...' or similar.
- **[writing]** Section 1 contains a paragraph in Chinese (lines 135-158) that appears to be a draft or translation artifact. This must be removed or fully translated into English to maintain consistency with the rest of the manuscript.
- **[writing]** In Section 4 (Scaling), the phrase 'MinT time-slices LoRA training sessions' (Section 4.2) and similar technical descriptions are clear, but the transition between the 'Scale Up' and 'Scale Down' subsections feels abrupt. Consider adding a brief bridging sentence to improve flow between these distinct scaling axes.
- **[writing]** The abstract uses the phrase 'MinT scales this adapter-revision path along three axes' followed by bolded terms. Ensure the subsequent sections (4.1, 4.2, 4.3) explicitly mirror this structure with clear topic sentences to maintain cohesion between the abstract and the body.